



ROBO RALLY

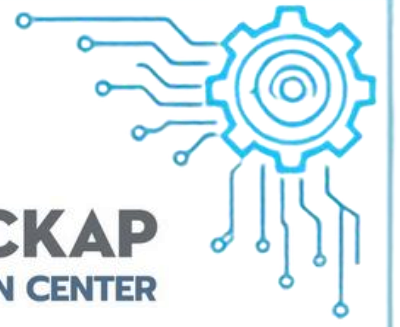
Robot Racing & Obstacle Challenge

"SPEED. SKILL. CONTROL. CONQUER THE TRACK."

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REGISTRATION FEES: Rs. 333/-
TEAM SIZE: MAXIMUM 3 MEMBERS
PRICE POOL: RS.12,750/-

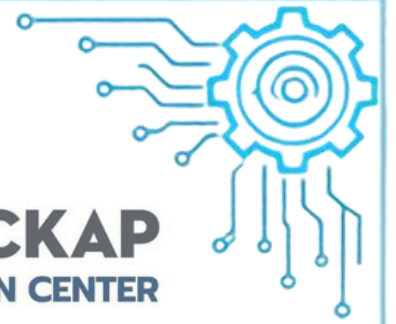
EVENT DESCRIPTION :

Navigate the Hurdles. Beat the Clock.

Put your robot's control, precision and agility to the test in **Robo Rally**. Participants must manoeuvre their bots through a mapped course filled with challenging hurdles and obstacles, tackling every section with accuracy and control. The goal is simple—complete the course in the shortest possible time. Every hurdle tests your skill. Every second counts.

Control. Navigate. Conquer.





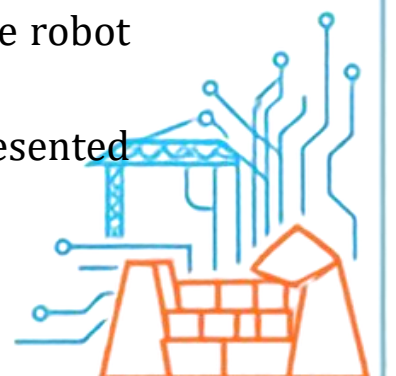
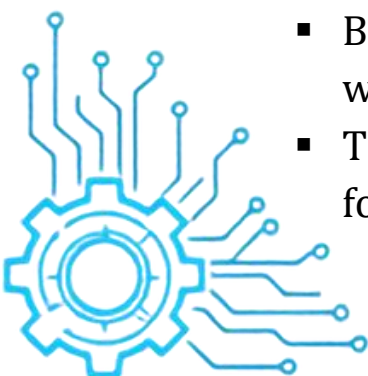
TECHNICAL INSPECTION & PRE-QUALIFICATION:

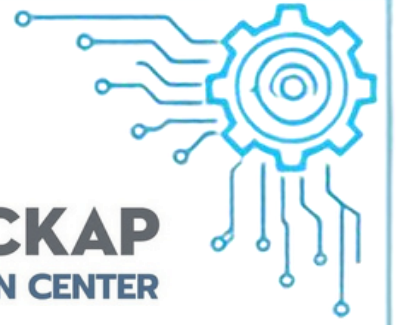
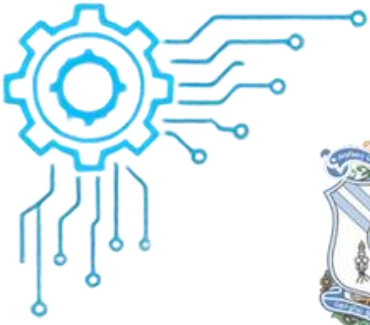
- All participating robots will undergo technical inspection before the commencement of the event.
- The robot presented for inspection must be the final competition-ready assembly.
- The robot must comply with all specifications provided by the organizers. The technical inspection will verify the robot's dimensions, weight, motor specifications and other submitted details.
- If the robot exceeds the specified maximum weight of 5 kg, it will be disqualified.
- If the motor operates above the specified maximum voltage of 16.8 V, the robot will be disqualified.
- If any other specification differs from the requirements provided by the organizers, the applicable penalty will be added.
- Only robots that successfully complete the technical inspection will be permitted to participate.
- Any modification to be made after inspection must be reported to the organizers. Unauthorized modification after inspection may result in penalty or disqualification.

REQUIRED DOCUMENTATION:

Each team must submit the following documentation during registration in the website

- Final Assembly – Clear representation/image of the final assembled robot.
- Robot Specifications – Complete technical specifications of the robot.
- Bill of Materials (BOM) – List of all components and parts used in the robot with cost estimation.
- The submitted documentation must correspond to the actual robot presented for inspection.





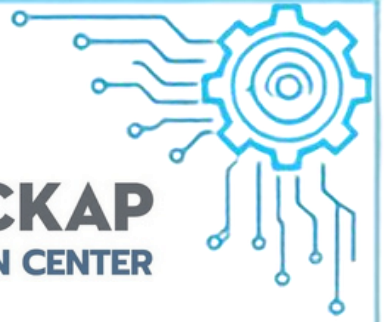
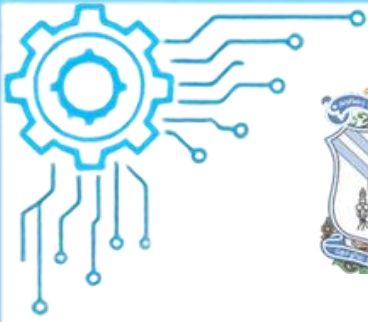
ROBOT SPECIFICATIONS:

- The robot must remain within the specified $25 \times 25 \times 20$ cm (L $\times$ B $\times$ H) dimensions.
- Only wireless robots are allowed
- The robot must not exceed the specified maximum 5kg weight. The maximum tire diameter is 10 cm, width 3 to 4 cm
- The motor voltage must not exceed 16.8 V.
- Any other specifications provided by the organizers must also be followed. The bot can have 2-wheel drive or 4-wheel drive.
- The robot body must not be taken from any ready-made toy. Robot-making kits can be used.
- IC engines are not permitted.
- The robot must be mechanically and electrically safe. wires should be properly sealed and enclosed

GENERAL RULES:

- Registration is online only. No offline or on-spot registration will be permitted.
- Each team can have a maximum of 3 members. Each member must carry their institute ID card. Each team is allowed to participate with one robot.
- Unethical behavior or intentional interference may lead to disqualification.
- The decision of the referee and organizers will be final.



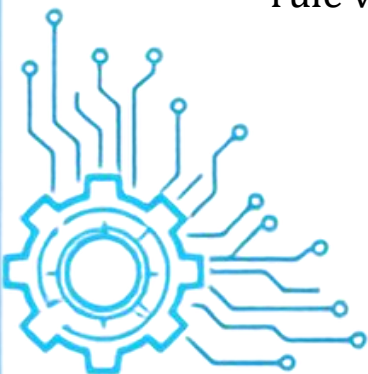


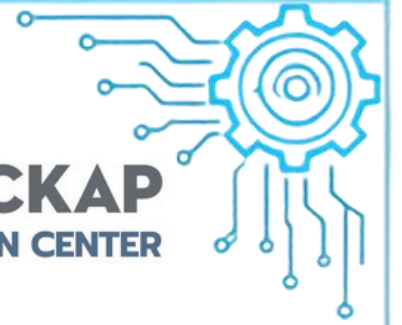
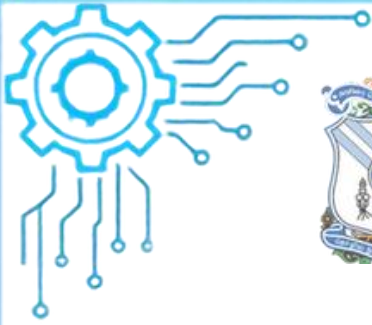
ARENA SPECIFICATIONS:

- The arena will consist of a 30-metre-long track with a width of 50 cm, containing specially designed obstacles and terrains.
- The arena will be prepared and controlled by the organizers. Participants are not allowed to modify or damage any part of the arena. Only authorized team members and coordinators are allowed inside the competition boundary.
- Arena specifications may be modified by the organizers if required.

PENALTY RULES:

- The winner is decided based on **fastest completion time + penalties**.
- The timer starts at the START signal and stops when the robot crosses the FINISH line.
- +5 seconds – Boundary touch.
- +10 seconds – Obstacle knocked down.
- +15 seconds – Hand touch/recovery.
- +20 seconds – Robot completely leaves the track.
- +25 seconds – Skipping an obstacle or checkpoint.
- Specification variation other than weight/motor voltage: Applicable penalty
Weight exceeding 5 kg: Disqualification
- Motor voltage exceeding 16.8 V: Disqualification
- Unauthorized assistance / intentional interference: Disqualification
Maximum 3 hand touches are allowed per run.
- The final score is calculated as: **Actual Time + Penalty Time = Final Time**. The team with the lowest final time gets the highest rank.
- If two teams have the same final time, a tie-breaker run may be conducted. Intentional rule violations or dangerous actions may result in disqualification





GAMEPLAY RULES:

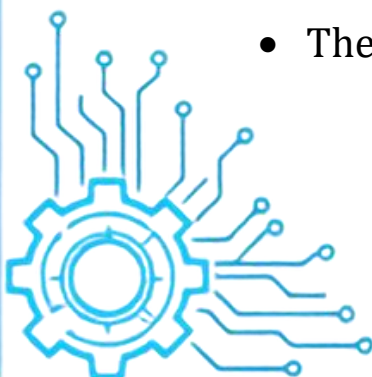
- Each qualified team will get one official attempt.
- The robot must navigate through the complete 30-metre track. The time taken to complete the track will be recorded.
- The robot must pass through the designated obstacles. Unauthorized human assistance during the run is not permitted. Any technical issue must be immediately reported to the referee. Participants must follow the instructions of the referee and event coordinators.

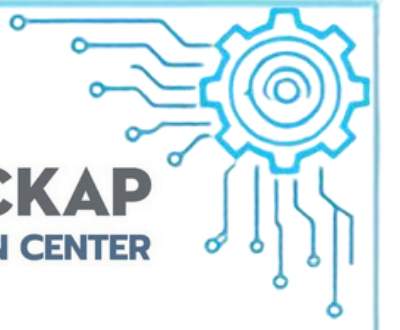
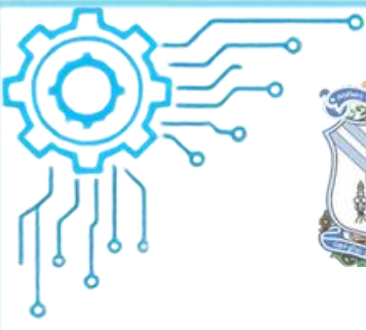
JUDGING CRITERIA:

- Each team's performance will be evaluated based on the time taken to complete the entire track and the penalty time incurred during the run.
- The recorded completion time and applicable penalty time will be combined to determine the team's Final Time.
- **Final Time = Completion Time + Total Penalty Time.** Teams will be ranked in ascending order of Final Time.
- The team with the lowest Final Time will be declared the Winner. The next lowest Final Times will determine the subsequent positions.
- A run will be considered valid only if the robot successfully completes the designated track while complying with the event rules.

REFEREE AND ORGANISERS:

- The referee's decision during the competition will be final.
- Organizers may modify or clarify the rules under special circumstances. Any changes will be communicated to the participating teams.
- The decision of the organizers will be final and binding.





NOTE:

- All parts and components of the robot must be clearly visible during technical inspection.
- Fully covered or enclosed assemblies that prevent the inspectors from verifying the components and specifications will not be accepted for inspection.

FOR FURTHER DETAILS CONTACT:

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